

Instructor: Diana Kazolias, M.S.,M.B.A.

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This introductory course in modern pharmacology is suitable for students interested in clinical medicine, biological science research, the pharmaceutical industry and development of modern therapeutic agents. The emphasis will be on the big picture of "therapeutic drugs" and to prepare students for future success in modern medicine, research, industry, or matriculation to health science programs.

This course covers different drug categories for major organ systems including the nervous system, the cardiovascular system, and the endocrine/reproductive system, as well as drugs for conditions like cancer, hypertension, and inflammatory diseases. Drugs will be studied from a systemic approach emphasizing the role of drugs in cell signaling, cell receptors, and effects on biochemical pathways.

Text: Pharmacology 8e, Hitner and Nagle, McGraw-Hill, 2022 with online access.

Objectives:

1. The student will become familiar with the different classes of drugs currently used today.
2. The student will understand drug mechanisms and how these mechanisms related to normative physiology and pathophysiology of the human body.
3. The student will understand how drugs are administered, distributed, and metabolized in the body.

Course Schedule

DUE DATE	TOPIC	CHAPTER(S)
May 30	Introduction, Pharmacokinetics, Geriatrics	1, 2, 3
May 31	Review of ANS	5
June 2	Sympathetic, Parasympathetic, ganglia	6, 7, 8
June 6	Relaxants and anesthetics	9, 10
June 8	Review of CNS, sedatives, antipsychotics	11, 12, 13
June 12	Antidepressants, stimulants, lithium	14
June 14	Psychotomimetics, antiepileptics, antiparkinson's	15, 16, 17
June 16	Anesthesia and analgesia	18, 19, 20
June 20	Review cardiac, treatment of heart failure	21, 22

June 21	Antiarrhythmics, antianginal drugs	23, 24
June 23	Diuretics and antihypertensives	25, 26
June 26	Coagulants, anticoagulants, antianemics	27, 30
June 28	Hypolipidemics and nutrition	28, 29
July 5	Respiratory drugs and antihistamines	31, 32
July 7	GI disorders, intestinal motility	33, 34
July 12	Intro to endocrine, steroids	35, 36
July 13	Gonadal, contraceptives, ED drugs	37
July 18`	Thyroid, parathyroid, bone degeneration	38
July 20	Pancreas and diabetes, pituitary	39, 40
July 25	Antibacterial, antifungal and antiviral drugs	41, 42
July 27	Antineoplastics, antiseptics	44, 45

See Canvas for course outlines, more grade information, and updates. Check announcements often!

Grading: 40% exams, 50% homework, 10% case study

Grade scale:

94-100	A	80-83.9	B-	64-66.9	D
90-93.9	A-	77-79.9	C+	60-63.9	D-
87-89.9	B+	70-73.9	C-	Below 60	F
84-86.9	B				

Details:

This course requires access through McGraw-Hill Connect, which is accessible via the *McGraw Hill Connect* tab within Canvas

Each chapter has a PowerPoint file. Please read the chapters before attempting the corresponding assignments.

Each chapter has online assignments accessible through your Connect site. Complete the online Smartbook assignment before attempting the assignment for that chapter. These assignments are open book and untimed. Please take the time to work through them as they are designed to help you learn the material. The due date for each assignment is listed on each assignment. You can complete as much or as little of an assignment in a sitting, close your work and come back to it. I recommend that you not work for more than 30 minutes without a short break. Do not wait until close to the due date of an assignment to complete it. Late assignments receive a 10% grade penalty for each day they are late.

There are two exams accessible through Connect. You will have a window of ten days to complete each exam, see dates on Canvas. These are multiple choice exams of 100 questions. When you log onto the exam, you will have 120 minutes to complete it. These are closed book, timed exams. No credit for work completed after the time is up.

A case study will be assigned by the end the second week. This is a written evaluation of a medical situation and a pharmacological solution. This assignment is a one to two page essay that is submitted through Canvas. A grading rubric will be posted under this assignment on Canvas. It is due on 7/29 at noon. NO CREDIT for late work.

Should you have individual learning needs, please contact me the first week of the term to discuss accommodation.

This course is designed to be completed online and as such there are no office hours or class meetings. Please feel free to contact me via email at any time. I have also set up a discussion board for you to correspond with your fellow students. I will monitor this site also as needed. If you need help either with technology or material, do not hesitate to ask as most problems can be easily solved if addressed early.

Academic Honesty

The Biology Program at UNH Manchester will strictly adhere to the University policy on academic honesty, as published in the UNH Student Rights, Rules, and Responsibilities Handbook (<http://www.unh.edu/vpsas/handbook/academic-honesty>). By turning in any piece of work in this course, you declare that you have read and understand the policy, and that you did not engage in any form of academic dishonesty as defined in the Handbook.

Plagiarism can take many forms, such as: submitting someone else's work - in whole or in part - as your own; collaborating on answers for individual assignments or allowing your own work to be used by another student; copying information from a web site or other text without proper documentation; buying a pre-written paper or lab report. Cheating is mainly concerned with copying on exams or in lab, bringing crib notes into an exam or referring to notes, or textbook or any other source such as a programmable calculator, tablet, or cell phone during an exam.

All electronic devices must be turned off and put away for the duration of the exam. Ear buds are not permitted. Any instances of cheating or plagiarism will result in consequences that can range from a failing grade on the assignment for all students involved to dismissal from the University, as defined in the UNH Student Rights, Rules, and Responsibilities Handbook.

Netiquette Guidelines

Netiquette is the socially and professionally acceptable way to communicate on the Internet. Please abide by these guidelines of "netiquette" when using online communication tools with your classmates and instructor.

In discussion boards and email messages: Identify yourself. Begin messages with a greeting and close with your name. Avoid sarcasm. It can be misinterpreted and cause hurt feelings. Keep the dialog collegial and professional. Some discussion topics may be controversial. Do not flame -These are outbursts of extreme emotion or opinion. Think twice before you submit a response. You cannot edit or delete your posts once they have been submitted. Do not use offensive language or profanity. Use clear subject lines for your posts. Do not use all caps. It is the online equivalent of YELLING! Use emoticons to clarify your emotions. They add context to your words that cannot be seen otherwise. :) Be forgiving. Anyone can make a mistake.

Center for Academic Enrichment (CAE):

The **CAE** is a resource for all students enrolled in **UNH Manchester** courses. Our mission is to help under-prepared students prepare, prepared students advance, and advanced students excel. Trained peer tutors and professional staff provide free assistance with reading and writing in all courses, as well as content in all math and science courses. Support for other content areas is dependent on the background and competencies of current peer tutor staff. ***All enrolled students are entitled to 1 hour of free one-to-one tutoring per week, per course.*** In addition, peer and professional tutors may run study groups for you class and provide drop-in math tutoring. Please visit us online at manchester.unh.edu/student/center-academic-enrichment to learn more about our services or to make an appointment with a tutor.

ADA Statement:

The University is committed to providing students with documented disabilities equal access to all university programs and facilities. If you think you have a disability requiring accommodations, you must register with the Student Accessibility Services (SAS) office. The Student Accessibility Coordinator at UNHM is Jenessa Zurek. Please reach out to the SAS office via email at jenessa.zurek@unh.edu for registration information and disability related questions.

For Emotional and Mental Health Services: In partnership with The Mental Health Center of Greater Manchester, UNHM offers consultation visits on a walk-in basis in Room 410H (located in the Student Services suites 4th floor). Services include: Free confidential screening & consultation with a licensed mental health clinician. Referrals to mental health or substance misuse treatment, and assistance in understanding how to afford additional treatment (with or without insurance!) or find free services.

Confidentiality and Mandatory Reporting: The University of New Hampshire at Manchester and its community are committed to assuring a safe and productive educational environment for all students and for the university. To this end, the university requires UNH faculty and staff members, hall directors, resident assistants (RAs), and other students employed in leadership, supervisory, and/or mentoring roles [such as Peer Assistant Leaders (PALs), Mentor Ambassadors for Transfer Engagement (MATEs) and the Center for Academic Enrichment (CAE) mentors and tutors] to report to the university's Title IX Coordinator (Donna Marie Sorrentino, dms@unh.edu, 603-862-2930/1527 TTY) any incidents of sexual violence, domestic violence, stalking and harassment shared by students regardless of whether it occurs on or off the Manchester campus. If you wish to speak to a confidential support service provider who does not have this reporting responsibility because their discussions with clients are subject to legal privilege, you can find a list of confidential crisis centers here. Please contact the crisis center that services the town you live in. For more information about what happens when you report, how the university considers your requests for confidentiality once a report is made to the Title IX Coordinator, your rights and report options at UNH (including anonymous report options) please visit <https://www.unh.edu/affirmativeaction/reporting-students>.

This syllabus is subject to change.